

10 CFR 50.73

RA22-025

May 27, 2022

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001LaSalle County Station, Unit 2
Renewed Facility Operating License No. NPF-11
NRC Docket No. 50-374Subject: Licensee Event Report 2022-002-00, Turbine Stop Valve Limit Switch
Failure due to Lubricant DegradationIn accordance with 10 CFR 50.73(a)(2)(i)(B), Exelon Generation Company, LLC
(EGC) is submitting Licensee Event Report (LER) Number 2022-002-00 for LaSalle
County Station, Unit 2.There are no regulatory commitments in this letter. Should you have any questions
concerning this report, please contact Mr. Dan Mearhoff, Regulatory Assurance
Manager, at (815) 415-2800.

Respectfully,

John Van Fleet Jr
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station

**LICENSEE EVENT REPORT (LER)**

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk at: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

LaSalle County Station, Unit 2

2. Docket Number

05000 - 374

3. Page

1 OF 4

4. Title

Turbine Stop Valve Limit Switch Failure due to Lubricant Degradation

5. Event Date

Month	Day	Year
3	24	2021

6. LER Number

Year	Sequential Number	Revision No.
2022	- 002 -	00

7. Report Date

Month	Day	Year
5	27	2022

8. Other Facilities Involved

Facility Name	Docket Number
NA	NA
Facility Name	Docket Number
NA	NA

9. Operating Mode

1

10. Power Level

100 percent

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ OTHER (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER****Licensee Contact**

CJ Smith, Operations Director

Phone Number (Include area code)

(815) 415-2200

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	TA	33	N007	Y	NA	NA	NA	NA	NA

14. Supplemental Report Expected☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month	Day	Year

6. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On 6/13/2020 during surveillances of the turbine stop valves (TSV), limit switch 2C71-N006G failed. The component was sent offsite for failure analysis. Component failure analysis was complete on 3/24/2021 and found only a small amount of grease on the internal components. It was determined the switch failed due to dried out and gritty grease, likely due to a missing boot which could have caused the limit switch to dry out. This established firm evidence that a past inoperability condition existed.

The component 2C71-N006G is the TSV-4 Closure Reactor Protection System (RPS) channel A2 scram limit switch. This device is required by technical specifications to support LCO 3.3.1.1 function 8 (RPS) and LCO 3.3.4.1 (End of Cycle - Recirculation Pump Trip or EOC-RPT). Each TSV position is monitored by two limit switches (for TSV-4, A2 and B2 channels). This condition results in the TSV-4 input to RPS A2 as well as one channel of the EOC-RPT trip being INOPERABLE. The appropriate LCO Conditions and Required Actions were entered and tracked. The RPS and EOC-RPT functions were maintained due to the redundant channels remaining OPERABLE. There were no reportability thresholds applicable to this issue at the time of the event, and there was no loss of safety function. The condition is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) for operation or condition prohibited by Technical Specifications.

Immediate corrective actions were taken to replace the component and restore the associated channel to OPERABLE.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 2	05000- 374	2022	- 002	- 00

NARRATIVE**Plant and System Identification**

LaSalle Country Station Unit 2 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power. The affected system was the main turbine system. The affected component was the turbine stop valve (TSV) limit switch 2C71- N006G. This is the limit switch for the TSV number 4 Closure Reactor Protection System (RPS) A2 channel.

Condition Prior to Event

Unit(s): 2 Date: March 24, 2021
Reactor Mode(s): 1 Mode(s) Name: Power Operation

Time: 00:00 CDT
Power Level: 100 percent

Description

On 6/13/2020 the limit switch on Unit 2 TSV-4 failed to actuate during surveillance testing. The limit switch is normally held in the actuated position during unit operation and only allowed to return to the relaxed, shelf state position, once per 6 months. The component (2C71-N006G) was replaced and sent offsite for component testing and failure analysis. The component failure analysis (documented in report dated March 24, 2021) verified the limit switch malfunction and determined the cause was attributed to dried out switch lubricant.

Cause

The failure analysis functional testing verified the limit switch malfunction. The switch cam required significantly more force to move than new switches. Only a small amount of grease was found on the internal components and significant wear was found on the components due to lack of lubrication. The shaft O-ring was intact, but severely hardened/embrittled due to aging.

Reportability and Safety Analysis

The component 2C71-N006G is the TSV-4 Closure RPS channel A2 scram limit switch. This device is required to be OPERABLE per LCO 3.3.1.1 function 8 (RPS) and LCO 3.3.4.1 (EOC-RPT). TSV-4 was fully closed during the surveillance, which should have resulted in the 2C71-N006G switch being open. Each TSV position is monitored by two limit switches (for TSV-4, A2 and B2 channels). This condition results in the TSV-4 input to RPS A2 as well as one channel of the EOC-RPT trip being INOPERABLE. The appropriate LCO Conditions and Required Actions were entered and tracked. The RPS and EOC-RPT functions were maintained due to the redundant channels remaining OPERABLE. There were no reportability thresholds applicable to this issue at the time of the event, and there was no loss of safety function.

Upon receipt of the failure analysis on March 24, 2021, the station was made aware that a past inoperability period existed from the test failure that occurred on June 13, 2020. The inoperability period was greater than allowed by TS LCO 3.3.1.1 and 3.3.4.1, as indicated below. This event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by the plant's TS.

- LCO 3.3.1.1 requires that the RPS instrumentation for each function is OPERABLE, and Condition A applies to one or more required channels being inoperable requires that the channel or its associated trip system is placed in trip within 12 hours. In this case, the station had exceeded the completion time of 12 hours, when the malfunctioning channel was found inoperable during June 2020 surveillance testing. In addition, LCO 3.3.1.1 Condition D was not entered for the Completion Time of Condition A not met.

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LaSalle County Station, Unit 2	05000- 374	2022	- 002	- 00

NARRATIVE

- LCO 3.3.4.1 requires that the EOC-RPT instrumentation is OPERABLE, which includes two channels of TSV closure and TCV fast closure or Minimum Critical Power Ratio (MCPR) limits are applied.

Corrective Actions

Corrective actions taken in response to the conditions were:

- A work order was completed to lubricate the switch, and to restore the associated channel to OPERABLE.
- The switch was replaced in the next outage.
- Performed component failure analysis at off-site testing facility.

Previous OccurrencesUnit 1 LER 373-2019-001-00

During quarterly surveillances of TSV limit switch 1C71-N006C, the component exhibited degraded performance on May 20, 2018 and a failure on September 9, 2018. The component was replaced; and subsequent component failure analysis determined the cause was attributed to degradation of the switch lubricant due to exposure from a high temperature environment, which established firm evidence that the sluggish performance exhibited in May 2018 was related to the failure in September 2018 and that a past inoperability condition existed. Immediate corrective actions were taken to replace the component and restore the associated channel to OPERABLE. Additional actions included performance of a causal investigation, identification of a new limit switch model that has greater heat tolerance, with a component replacement schedule starting in 2019.

Unit 1 LER 373-2019-002-00

On March 23, 2019, during quarterly scram functional surveillances of turbine stop valves (TSV), the limit switch associated with TSV number 3 failed to return to shelf state when the lever arm was released. The failed component was replaced and sent offsite for component testing and failure analysis. The component failure analysis completed on May 3, 2019 determined the cause was attributed to insufficient lubrication in the shaft to bushing section of the limit switch. This established firm evidence that the failure mechanism and past inoperability condition may have existed since its last successful surveillance on December 15, 2018. Immediate corrective actions were taken to replace the component and restore the associated channel to OPERABLE. Additional actions included performance of a component failure analysis and to complete scheduled limit switch replacements of the current model with a new limit switch model that has greater heat tolerance.

Component Failure Data

Manufacturer: NAMCO Controls Corporation

Device: Unit 2 Turbine Stop Valve limit switch (Model: EA700-80154)

Component ID: 2C71-N006G, Unit 2 Turbine Stop Valve Position Switch (Cat. ID 0001791228)